Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	To be assigned
Project Name	Global Fuel Prices Analysis
Maximum Marks	4 Marks

Data Flow Diagrams

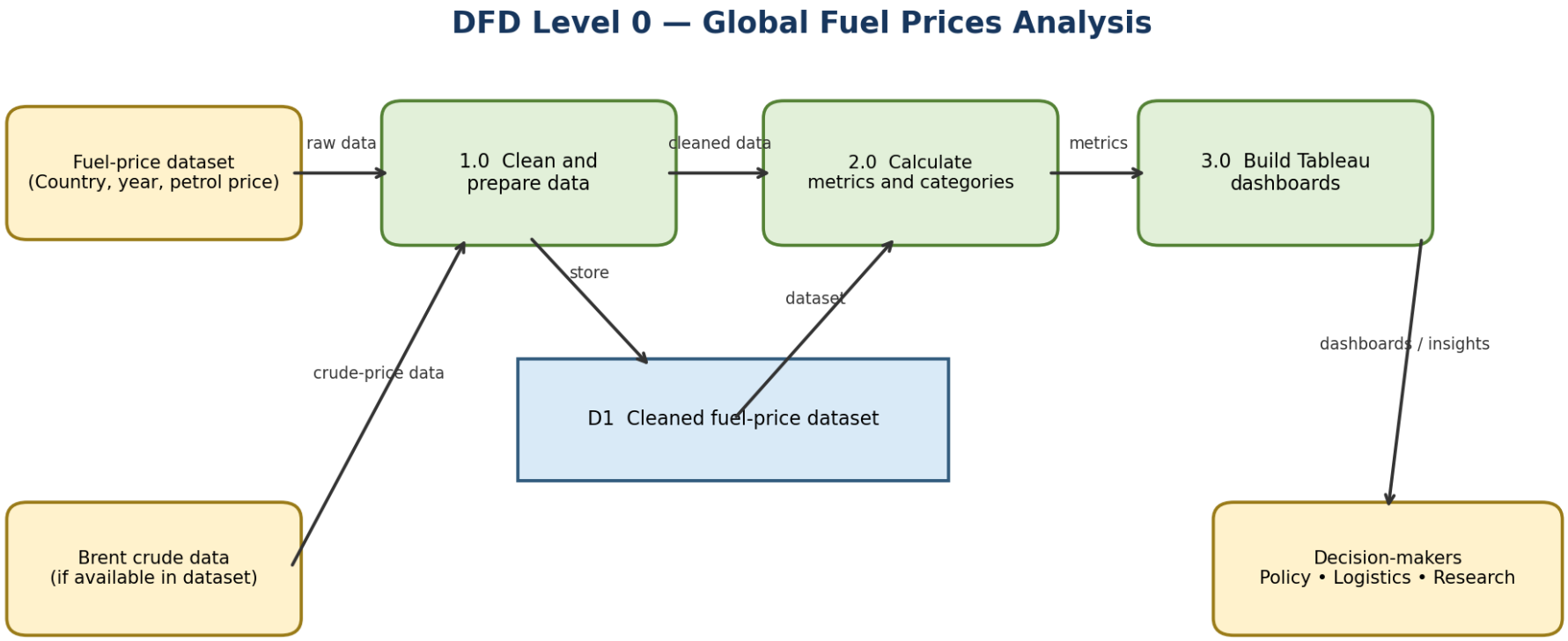
A Data Flow Diagram (DFD) shows how data enters the system, is processed and stored, and is delivered as useful information to its users. For this project, fuel-price data is cleaned, analyzed through calculated fields, and presented in interactive Tableau dashboards.

DFD Symbol Legend

Symbol	Name	Description
Oval / rounded shape	External Entity	A person or system outside the project that provides or receives information.
Numbered process box	Process	An activity that transforms input data into output information.
Data-store box	Data Store	A place where cleaned data or project information is stored.
Labeled arrow	Data Flow	Movement of data between entities, processes, and data stores.

DFD Level 0 — Global Fuel Prices Analysis

The diagram below shows the main data flow from source data through cleaning and analysis to Tableau dashboards used by policy makers, logistics managers, and researchers.



Data flow: source data → cleaning → analysis/calculations → Tableau dashboards → stakeholder insights

Key outputs: country-wise price comparisons, tax bands, price categories, available petrol-to-Brent comparisons, and dashboard insights. Results depend on the fields and years available in the supplied dataset.

User Stories

User stories describe the needs of people who use or maintain the Global Fuel Prices Analysis project. Priorities and sprint releases are proposed for this project plan.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Policy Maker	Dashboard & Policy Analysis	USN-1	As a policy maker, I can compare petrol prices and tax levels across countries.	Dashboard shows country-wise prices and tax comparisons with clear labels.	High	Sprint-1
Policy Maker	Price Comparison	USN-2	As a policy maker, I can identify countries with high and low petrol prices.	I can sort or filter countries and identify price differences.	High	Sprint-1
Logistics Manager	Cost Planning	USN-3	As a logistics manager, I can review fuel-price patterns to support transport cost planning.	Dashboard presents comparable prices and trends for selected countries and years.	High	Sprint-1
Researcher / Analyst	Trend Analysis	USN-4	As a researcher, I can explore fuel-price changes over the selected years.	Year and country filters update the relevant charts correctly.	High	Sprint-1
Researcher / Analyst	Brent Crude Comparison	USN-5	As an analyst, I can compare petrol prices with Brent crude prices where data is available.	The dashboard displays the comparison metric and clearly identifies available data.	Medium	Sprint-2
All Stakeholders	Interactive Filters	USN-6	As a dashboard user, I can filter visualizations by country, year, or price category where available.	All related charts respond to the selected filters without misleading labels.	High	Sprint-1
All Stakeholders	Dashboard Navigation	USN-7	As a dashboard user, I can navigate the Policy, Logistics, and Research dashboards.	Each dashboard opens and its purpose and visualizations are understandable.	Medium	Sprint-2
Project Administrator	Data Quality	USN-8	As a project administrator, I can use a cleaned dataset with consistent fields and formats.	Missing or inconsistent values are handled and field names are consistent.	High	Sprint-1
Project Administrator	Calculated Fields	USN-9	As a project administrator, I can use calculated fields for tax band, price category, and price-Brent ratio.	Calculated fields follow the documented rules and display valid results for available data.	High	Sprint-1
Project Administrator	Publishing & Access	USN-10	As a project administrator, I can publish the Tableau story for stakeholders to view.	The published story link opens and the dashboards are viewable online.	High	Sprint-2
Researcher / Analyst	Insights & Interpretation	USN-11	As a researcher, I can use charts to identify patterns and differences across regions.	Charts include titles, units, legends, and readable comparisons.	Medium	Sprint-2
All Stakeholders	Documentation	USN-12	As a stakeholder, I can understand the data source, scope, and limitations of the analysis.	Project documentation states the dataset scope and explains key measures and limitations.	Medium	Sprint-2